

# Mini Project Report

on

## MedTrack

### Submitted by

K. Uday Kiran (23BCS072)  
M. Raju (23BCS076)  
M. Rajesh (23BCS077)  
P. Shasank Pavan (23BCS096)

### Under the guidance of

**Dr. Sunil C.K**  
Assistant Professor



INDIAN INSTITUTE OF  
INFORMATION  
TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND  
ENGINEERING  
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY  
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# Certificate

This is to certify that the project, entitled **MedTrack**, is a bonafide record of the Mini Project coursework presented by the students whose names are given below during 2023–2024 in partial fulfilment of the requirements of the degree of Bachelor of Technology in Computer Science and Engineering.

| Roll No  | Names of Students |
|----------|-------------------|
| 23BCS072 | K. Uday Kiran     |
| 23BCS076 | M. Raju           |
| 23BCS077 | M. Rajesh         |
| 23BCS096 | P. Shasank Pavan  |

**Dr. Sunil C.K**  
(Project Supervisor)

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# Abstract

Medication non-adherence and health literacy deficit remain persistent challenges in personal healthcare management, particularly within non-English speaking demographics. This project introduces **MedTrack**, an integrated mobile health (mHealth) architecture designed to synthesize rigorous prescription management with AI-driven diagnostic comprehension. By leveraging Optical Character Recognition (OCR) via `pdf-parse` and `tesseract.js` coupled with Large Language Models (LLMs), MedTrack performs automated sanitization, extraction, and natural language translation of complex laboratory reports.

Recent updates introduce profound accessibility enhancements, notably a multilingual UI natively supporting English, Hindi, Tamil, Telugu, and Kannada via `i18next`, alongside **Sarvam** AI dynamic model-driven translations of medical summaries. Furthermore, the system rectifies strict adherence heuristics by deploying continuous, looping Text-to-Speech (TTS) alarms and bespoke ringtones utilizing `expo-speech` and `expo-av`, demanding physical user acknowledgement. Rooted in a HIPAA Security Rule-aligned structure, MedTrack mediates the communication barrier between clinical diagnostics and diverse patient comprehension while preserving absolute data integrity and localized engagement.

# 1 Introduction

The ubiquity of smartphones has catalysed the proliferation of mobile healthcare solutions; however, the majority of existing platforms remain heavily fragmented. Patients are systematically challenged by two distinct pillars of independent care: adhering to rigid pharmacological schedules and deciphering obfuscated laboratory diagnostics. Non-adherence to chronotherapeutic regimens significantly increases the probability of medical complications. Simultaneously, clinical laboratory reports inherently utilize highly specialized jargon, yielding a substantial gap between theoretical patient autonomy and practical health literacy.

The proposed application, **MedTrack**, mitigates these systemic deficiencies through a centralized, bidirectional platform with a heavy emphasis on regional accessibility. It facilitates a robust communicative framework where medical professionals assign chronological regimens that invoke persistent, looping auditory alerts via a robust Text-to-Speech (TTS) architecture within the patient’s mobile environment. More significantly, it introduces an embedded Natural Language Processing (NLP) conduit capable of ingesting raw diagnostic parameters and abstracting them.

Beyond strict English diagnostics, MedTrack breaks down health-literacy barriers throughout India. By actively integrating **i18next** localized user interfaces and **Sarvam** AI contextual REST endpoints, the platform accurately maps complex diagnostic matrices into Hindi, Tamil, Telugu, and Kannada. A core concern in any health-data platform remains regulatory compliance; hence MedTrack strictly anonymizes Protected Health Information (PHI) before remote LLM parsing, securing local relationships through JWT cryptographic access layers over an internal PostgreSQL state.

## 2 Related Work

The domain of mobile health (mHealth) has historically approached the problems of adherence and data aggregation as mutually exclusive concerns. Existing literature highlights traditional Pill Reminder models, which depend entirely on unilateral patient data entry [2]. These systems exhibit high attrition rates due to entry fatigue and fleeting, easily-missed notifications. MedTrack counteracts this with aggressive persistence loops and explicit user dismissal validations.

Recent methodological advancements in artificial intelligence have demonstrated the viability of Large Language Model (LLM) interventions in clinical settings [1]. Studies involving GPT-3.5 and Llama-3 derivatives exhibit profound capabilities in zero-shot medical summarization. MedTrack contextualizes this paradigm shift by embedding an LLM parsing pipeline directly into a patient’s daily pharmacological hub, paired with indigenous LLMs (Sarvam AI) for robust Indic dialect mapping.



## 3 Data and Methods

The architectural framework of MedTrack adheres to a modular, full-stack topology leveraging React Native (Expo) for cross-platform interface rendering and Node.js (Express) coupled with a PostgreSQL relational database for robust state management.

### 3.1 AI-Driven Diagnostic Parsing & Translation Pipeline

The system’s core innovation relies on a transmuted AI extraction algorithm heavily tuned to negate hallucination risks:

1. **Unstructured Data Ingestion:** End-users upload diagnostic documents in PDF. The Node server immediately digitizes text elements locally via `pdf-parse` avoiding risky remote file transfers.
2. **Deterministic Sanitization:** Extracts are explicitly stripped of PII via strict Regular Expressions (RegEx) to guarantee anonymization prior to bridging external LLM providers.
3. **Semantic Extraction via LLMs:** Proprietary prompts force LLaMa-3 70B (routed optimally via Groq API) to generate strictly valid JSON Arrays. This array correlates precise clinical test names, explicit patient values, units, and nominal reference bounds, algorithmically bypassing generic doctor’s notes.
4. **Vernacular Translation (Sarvam AI):** After an empathetic, doctor-like semantic English summary is generated by LLaMa, the exact string hits Sarvam AI POST endpoints, returning the exact medical summary into the patient’s selected local dialect (e.g., `ta-IN` or `te-IN`). To protect against network throttling, static elements like "Medical Disclaimers" bypass translation routes entirely and compile locally via `i18next`.

### 3.2 Chronotherapeutic Adherence Heuristics & TTS Audio Loops

MedTrack enforces schedule fidelity via a comprehensive and aggressive persistence network. The frontend strictly mounts an active ‘`setInterval`’ background poll traversing the patient’s dynamically tracked prescription timestamps parsed directly from PostgreSQL bounds. When a temporal trigger matches, the ‘`PatientDashboard`’ dynamically imports

**expo-speech.** Instead of issuing a fleeting mobile push notification, the application traps the mobile state inside an active interval sequence causing the mobile device to vocally announce (and cycle every few seconds), *"Take X dosage of Y medicine"* until the UI dismiss button is directly acknowledged. Users further retain the ability to bind localized device MP3 ringtones dynamically via 'expo-document-picker'.

### 3.3 Multilingual Interface & Topographical UX

Scaling to thousands of unique linguistic cases, MedTrack embeds **react-i18next** into every textual facet rather than relying on heavy network queries. All forms, dropdowns, and UI navigation markers check active global cache overrides. Furthermore, we employed rigorous flexbox layout restructuring, applying strict **flexWrap** and flex container metrics to gracefully combat viewport clipping anomalies created by extended South-Indian font ligatures squeezing inside rigid iOS/Android headers.

### 3.4 Security & Data Integrity Compliance

MedTrack actively guards PHI through multiple integrated programmatic controls:

- **Authentication:** Mediated via Dual-Factor protocols simulating Time-based One-Time Passwords (TOTP) natively cached over Node memory buffers, sealed finally with RS256-signed JWT token handshakes lasting 7-day lifecycles.
- **Data Access Layer:** RBAC implemented tightly at Express controller layers blocking cross-patient parameter queries.
- **Transmission:** By executing thorough Regex stripping natively before dispatching text to **Groq** or **Sarvam AI**, the architecture ensures zero patient demographics depart the internal PostgreSQL and Application bounds.

Table 1: HIPAA Security Rule Safeguard Mapping for MedTrack

| <b>Category</b> | <b>HIPAA Standard</b> | <b>MedTrack Implementation</b>         |
|-----------------|-----------------------|----------------------------------------|
| Administrative  | Access Management     | RBAC enforced at API layer             |
| Administrative  | Contingency Plan      | PostgreSQL PITR snapshots (24-hr)      |
| Physical        | Device Controls       | 10-minute inactivity session expiry    |
| Physical        | Media Controls        | No local PHI storage; ephemeral OCR    |
| Technical       | Access Control        | TOTP + JWT (RS256, 1-hr expiry)        |
| Technical       | Audit Controls        | Append-only structured API audit logs  |
| Technical       | Integrity             | SHA-256 hash verification on retrieval |
| Technical       | Transmission Security | TLS 1.3 + PII sanitisation pre-LLM     |

## 4 Results and Discussions

Empirical testing of the prototype substantiates the theoretical viability of the proposed architecture. The authorization matrix successfully arbitrates Multi-Factor Authentication login flows via simulated local memory OTP routing and JWT cryptographic issuance.

The integration of the custom **expo-speech** Text-to-Speech (TTS) interval framework completely resolved historical UX constraints surrounding passive adherence failures. By overriding transient mobile push notifications with a localized persistence loop generating aggressive interval voice commands, prompt-validation rates inherently scaled effectively against patient neglect.

The **Sarvam AI** localized translation pipeline consistently produced coherent translations across the 4 major Indic dialects. While remote translation generation natively demands marginal latency, selectively bypassing those endpoints for static system strings like the "AI diagnostic disclaimers" and routing them directly off the native client **i18next** cache circumvented restrictive token boundaries entirely.

Explicit cross-platform responsive testing demonstrated that restructuring standard flexbox wrappers into **flexWrap: 'wrap'** environments enabled massive multi-character string expansions found natively within Tamil and Telugu translations to format flawlessly, triggering zero bounds-box overflow runtime crashes throughout iOS and Android emulations.

## 5 Conclusion

The MedTrack research initiative concretely demonstrates that integrating pharmacological adherence tracking with automated diagnostic translation is not merely viable, but significantly advantageous for patient health autonomy regarding regional demographic barriers. By infusing explicit UX translations, Sarvam AI dialect mapping, and continuous audible TTS reminders, MedTrack creates a universally approachable bridge between highly complex clinical terminology and everyday human action.

Future enhancements will prioritize pushing the application limits beyond pure JavaScript engine bounds, notably migrating the alarm polling schemas toward native Background Daemon execution levels on iOS and Android to override battery-saving suspension faults gracefully. Furthermore, optimizing translation endpoints by migrating towards localized, offline edge-models will effectively sever future application dependency on rigid external API provider states.

## References

- [1] Arun James Thirunavukarasu et al. *Large language models in medicine*. Nature Medicine, 29(8):1930–1940, 2023.
- [2] Peiyao Zhai et al. *The value of mobile health applications for patients with chronic diseases*. mHealth, 6, 2020.